Gravity above the surface

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Learning target: Use distance from a planet’s center to compare fields at altitude.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 4.*
3 N; divide by 25 because force follows inverse square separation.

02. Worked example: explain the model
Radius R plus altitude 2 R makes 3 R; using h alone gives the wrong denominator.

03. Read and interpret the model
h=R; center separation r=2 R.

04. Apply the physics
3 N/kg.

05. Explain your reasoning
Astronaut and spacecraft fall together; little support force. Floating is not evidence of absent gravity.

06. Review: connect an earlier idea *Review: Unit 1, Day 17.*
At rest: friction 24 N opposite the push. Sliding: net force 24 N with the push.

07. Exit check
1 N/kg.

08. STRETCH: test the model
Center distance doubles to 2 R, so altitude is R.

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01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 4.*
Fixed masses: 3 r. Both masses enlarged: 15 r gives the required force fraction.

02. Worked example: explain the model
Radius R plus altitude 2 R makes 3 R; using h alone gives the wrong denominator.

03. Read and interpret the model
h=R; center separation r=2 R.

04. Apply the physics
8.68 N/kg; 2.45 N/kg.

05. Explain your reasoning
Astronaut and spacecraft fall together; little support force. Floating is not evidence of absent gravity.

06. Review: connect an earlier idea *Review: Unit 1, Day 17.*
N=39.2 N; static limit 19.6 N. Low push: friction 8 N, a=0. High push: kinetic friction 15.68 N, a=2.08 m/s².

07. Exit check
h=R; h=2 R.

08. STRETCH: test the model
Center distance r/R=√10; altitude h/R=√10-1≈2.16.